

HARSHAVARDHAN REDDY NARRA

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EDUCATION

University of Southern California (USC), Los Angeles, CA

Aug 2025 – Dec 2027 (Expected)

M.S. Electrical and Computer Engineering — VLSI and Computer Architecture

- **Completed:** EE457 Computer Systems Organization; EE477 MOS VLSI Circuit Design. **Fall 2026:** EE557 Computer Systems Architecture (Pin branch predictors); EE511 Machine Learning Hardware Accelerators (LeNet-5, activation PCA/SVD)

Jawaharlal Nehru Technological University Hyderabad, India

Aug 2021 – May 2025

B.Tech. Electrical and Electronics Engineering

TECHNICAL SKILLS

Languages: C, C++, Python, CUDA, Verilog, SystemVerilog, TCL

Simulation: QuestaSim, ModelSim, Vivado, Vitis HLS, Cadence Virtuoso, Intel Pin

Concepts: Computer architecture, RTL design, FPGA prototyping, RISC-V, AMBA AXI4, systolic arrays, branch prediction, ML accelerators, GPU architecture, PCA/SVD compression, CMOS / full-custom VLSI

Tools: Linux, Git, PyTorch, NumPy, AWS, Zynq-7020, Kintex-7

EXPERIENCE

Research Intern, Indian Institute of Technology (IIT) Bhubaneswar

Jan 2026 – Present

Streaming 1024-bit hyperdimensional computing accelerator · Advisor: Prof. Srinivas Boppu

- Designed synthesizable SystemVerilog RTL for a 1024-bit AI classifier (XOR bind, permute, bundle, Hamming/popcount) and a Python golden model; bit-exact on Zynq-7020 at 100 MHz.
- Measured 493,512 EMG windows on hardware: **72.78%** accuracy, **4.63 μ s / 12.0 μ J** per window vs ARM 818 μ s / 2.09 mJ (**$\sim 177\times$ faster, $\sim 174\times$ more energy-efficient**).
- Brought up scatter-gather DMA from DDR (~ 216 k windows/s) so the processor does not copy each transfer.

Teaching Assistant, University of Southern California (USC)

Aug 2026 – Present

TAC 348: Making Smart Devices · Prof. Rob Parke

- Led weekly embedded-systems labs covering Particle Photon 2 firmware, GPIO, I2C sensors, finite-state machines, Bluetooth/Wi-Fi, and cloud IoT (Blynk, REST APIs, dashboards).
- Mentored hardware–software bring-up and debugging: firmware, sensor interfaces, communication, and circuit faults on the environment-monitor, Bluetooth-car, and smart-watch builds.
- Guided students through end-to-end smart-device prototypes, from peripheral interfacing and embedded control logic to wireless links and cloud-connected applications.

SELECTED HARDWARE PROJECTS

Systolic-Array CNN Accelerator for Vision Transformer Feature Extraction

Verilog · Kintex-7 FPGA

- Designed a 5×6 systolic PE array (convolution, pool, activation) as CNN IP on a Vega AS1061 RISC-V host over AXI4.
- Mapped ViT feature extraction onto the array: **116 GOP/s at 200 MHz**, **$2.8\times$** lower stage latency vs software; 2.498 W on Genesys 2. IEEE HiPC 2024 paper; First Runner-Up, DVCon India 2024 (team of 3).

Branch Predictors with Intel Pin

EE557

- Implemented always-taken, 2-bit global, 2-bit bimodal (PC[4:0]), and 2-bit correlated (4-bit GHR + PC[3:0]) predictors as C++ Pin analysis routines.
- Instrumented **tar** on **9.70M** branches: always-taken **74.3%**, 2-bit global **79.3%**, bimodal **80.0%**, correlated **82.1%**.

LeNet-5 Training and On-Chip Activation PCA

EE511 · PyTorch

- Coded LeNet-5 from primitive layers (no model zoo): MNIST test accuracy **99.06%**; Fashion-MNIST **89.58%**; layer-wise MAC / parameter / intensity table.
- Wrote PCA/SVD from scratch on activations; rank-16 SVD on C5/F6 cut $\sim 80\%$ of those weights within 1 point of the uncompressed baseline.

5-Stage Pipelined RV64I Processor

LFX Mentorship

- Implemented an in-order RV64I + Zba Harvard pipeline in SystemVerilog (forwarding, load-use stall, branch flush) with a self-checking testbench driven by compiled C.

HONORS, COMPETITIONS & CERTIFICATIONS

- **First Runner-Up**, DVCon India 2024 Design Contest; peer-reviewed author, **IEEE HiPC 2024** (ROCS).
- **Ranked #1**, EE477 MOS VLSI Circuit Design (USC); LFX Mentorship coding challenge (RISC-V).
- **Cadence:** SystemVerilog for Design and Verification (v25.03), Apr 2026; Essential SystemVerilog for UVM, May 2026.